QISKIT FALL FEST 2026 × IIIT DELHI | TRACK 2: QUANTUM OPTIMIZATION

XD CLAN

QAOA for Constrained Portfolio Optimization

From a financial selection problem to a binary model, a quantum circuit, and evidence-based comparison.

6-ASSET CASE

64 possible bitstrings
20 feasible portfolios

12-ASSET CASE

924 feasible portfolios
exact classical reference

RECORDED HARDWARE RUN

IBM Marrakesh · 1,024 shots

LIVE OPTIMIZER

https://qiskit-fall-fest-2026-iiitd-ajiq5qdibj2ujasptcycxm.streamlit.app/

Updated explanatory presentation · figures grounded in saved project CSVs

What does portfolio optimization mean?

The app and notebook treat portfolio selection as a discrete decision problem.

DECISION

Each asset is a binary choice:
1 = include the asset
0 = leave it out

Portfolio = vector x of 0s and 1s.

TRADE-OFF

Expected return rewards attractive assets.
Risk penalizes portfolios with correlated or volatile assets.

CONSTRAINT

Choose exactly K assets.
Advanced benchmark also limits how many assets can come from each sector.

Why it is hard: with n assets there are 2^n binary portfolios. Constraints reduce the valid set, but the search space grows quickly.

Project objective: minimize a seeded risk/return cost while respecting the portfolio constraints.

How the portfolio score is built

The experiment uses a seeded return vector and a positive-semidefinite covariance-like matrix.

OBJECTIVE

$C(x) = q\, x^T \Sigma x - \mu^T x$
Risk term − return term

CARDINALITY PENALTY

Add $\lambda(\Sigma_i x_i - K)^2$ to penalize selecting too few or too many assets.

Inputs

μ: expected returns
$\Sigma = AA^T$: risk structure

Binary choice

$x_i \in \{0,1\}$
select K assets

Cost evaluation

compute risk/return
plus constraint penalty

Lower cost is better in this formulation. The random seed makes the recorded test instances reproducible.

From a user problem to a measured result

This is the pipeline used by the project; simulator and hardware branches are reported separately.

1. Define

assets, returns, risk matrix, K

2. Encode

cost Hamiltonian + penalty

3. Optimize

QAOA angles γ and β

4. Sample

measure bitstrings and count shots

5. Evaluate

feasibility, cost, classical comparison

SIMULATION BRANCH

Statevector sampling and synthetic noisy Aer tests.

REAL HARDWARE BRANCH

Recorded IBM Quantum Runtime job on ibm_marrakesh.

Important: the 8-qubit noisy pilot is a simulation; the saved 6-qubit run is the documented real-QPU experiment.

QAOA alternates cost and mixing operations

The circuit parameters are tuned by a classical optimizer using measured or simulated objective values.



Depth p

p counts the number of cost/mixer layers. Larger p adds expressive power and circuit cost.

Important nuance

More depth is not guaranteed to improve a finite-shot run; results depend on optimization and sampling.

A good score is not enough: the portfolio must be feasible

For the small benchmark, exactly 3 of 6 assets must be selected.

Asset 1	Asset 2	Asset 3	Asset 4	Asset 5	Asset 6
SKIP	SELECT	SELECT	SKIP	SKIP	SELECT

Exact optimum: assets 2, 3, 6 • cost = −0.5030146903

Feasibility

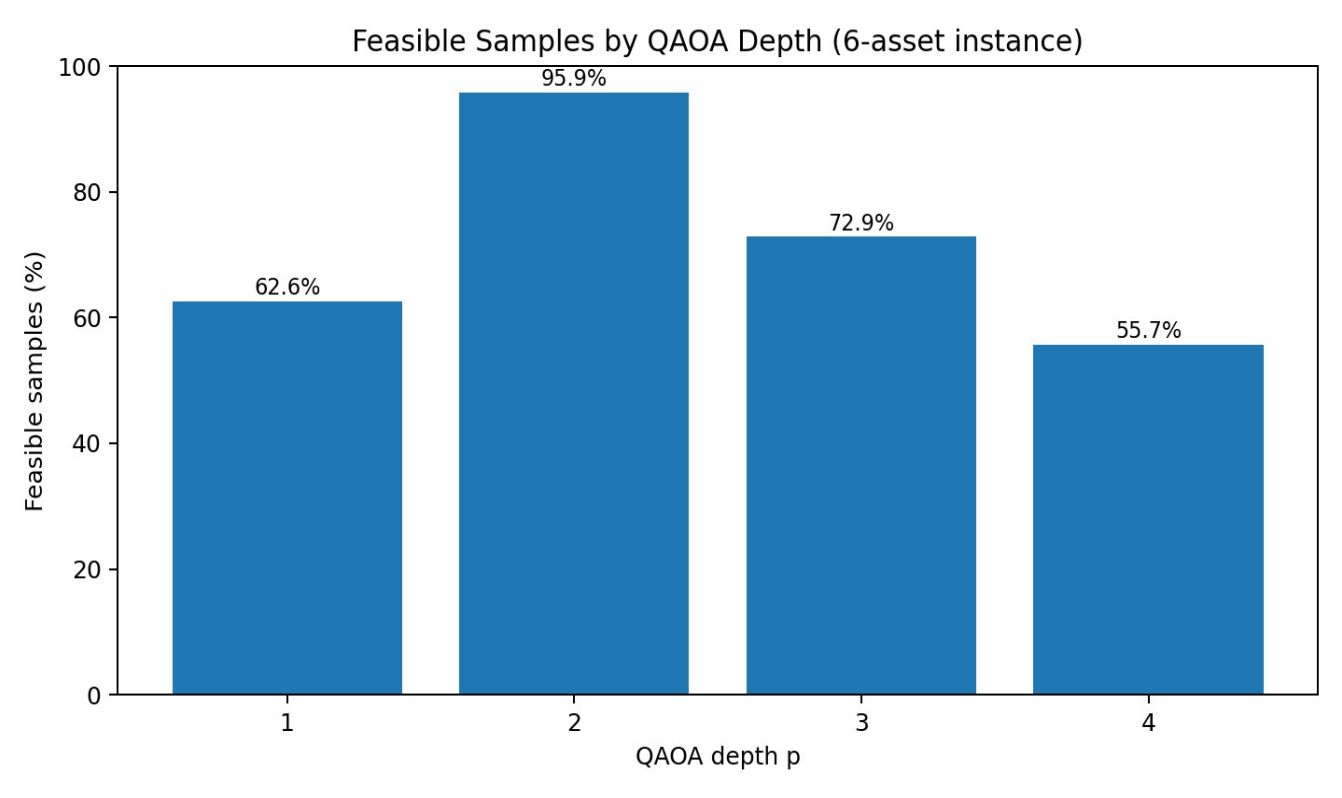
Exactly K assets selected; otherwise the candidate violates the cardinality constraint.

Penalty λ

A penalty makes invalid portfolios costly, but too small or too large a λ can change sampling behaviour.

Feasibility varies with QAOA depth

Each depth has 4,096 recorded simulator shots; a feasible sample selects exactly three assets.



BEST FEASIBLE RATE

p = 2: 95.87% of samples were feasible in this recorded run.

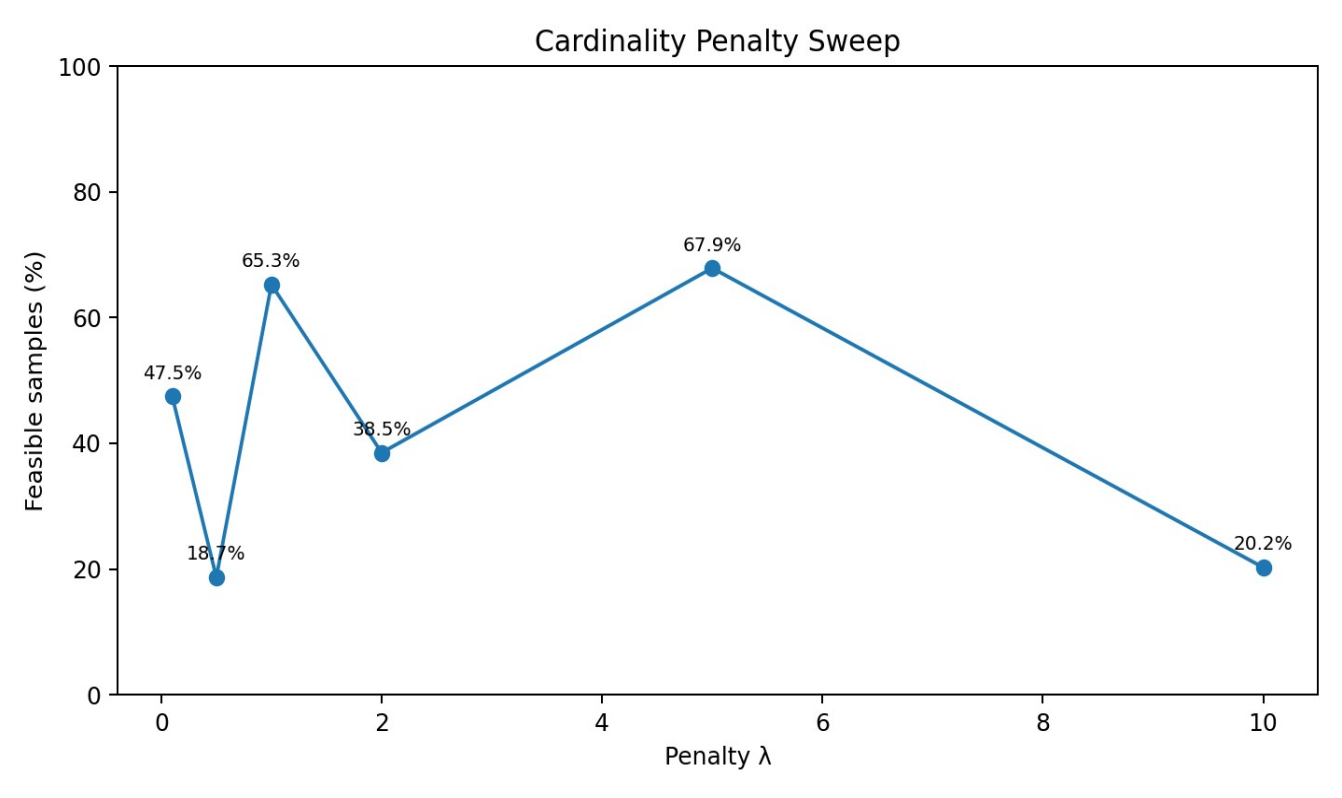
BEST-PORTFOLIO HIT RATE

p = 2: 183 / 4,096 shots = 4.47% matched the exact optimum.

This is a finite-shot simulator result, not a hardware comparison.

Penalty strength changes the feasible-sample rate

Recorded 4,096-shot evaluations for each tested λ.



HIGHEST OBSERVED

λ = 5.0 → 67.87% feasible samples in this sweep.

NOT MONOTONIC

λ = 0.5 gives 18.68%; λ = 10 gives 20.24%. Bigger penalty does not automatically mean better sampling.

A parameter sweep is empirical: it identifies promising settings for this instance, not a universal rule.

Exact classical search gives a ground-truth reference

The 12-asset case chooses 6 assets; there are 924 feasible combinations.

EXACT SEARCH

924 feasible portfolios
fully enumerated

OPTIMUM COST

−0.6872192458
exact classical reference

QAOA p = 1

4,096 shots
exact optimum appeared once

Optimal assets: 1, 4, 6, 8, 9, 12

Why keep the classical baseline? It tells us whether a sampled candidate is optimal, how far it is from the best-known cost, and whether the quantum heuristic adds value on the tested instance.

The 20-asset result is a classical exhaustive benchmark

It shows how the constraints can be checked at a larger portfolio size without mislabeling it as a quantum result.

SEARCH SPACE

184,756 combinations
choose 10 from 20

SECTOR-FEASIBLE

80,000 portfolios
passed the configured cap

BEST COST

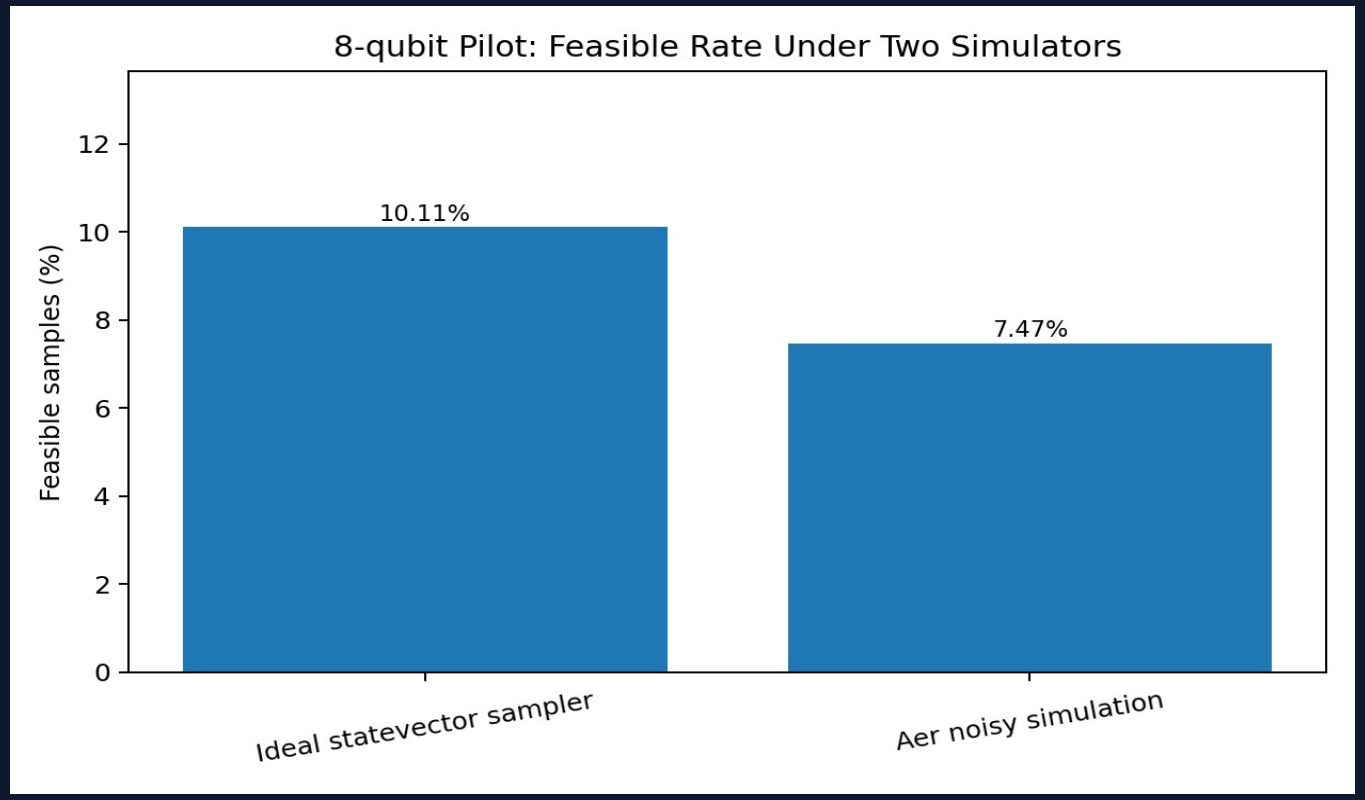
−1.4913972690
classical exhaustive search

Recorded best portfolio: [1, 2, 5, 6, 7, 8, 12, 14, 15, 20]

This is not a 20-qubit QAOA experiment. Keeping the label explicit makes the evidence more credible and prevents overstating what the quantum circuit actually solved.

Synthetic noise reduced the pilot’s feasible rate

8-qubit pilot, 2,048 shots per simulator; noise settings are illustrative, not calibrated to a device.



IDEAL SAMPLER

10.11% feasible samples

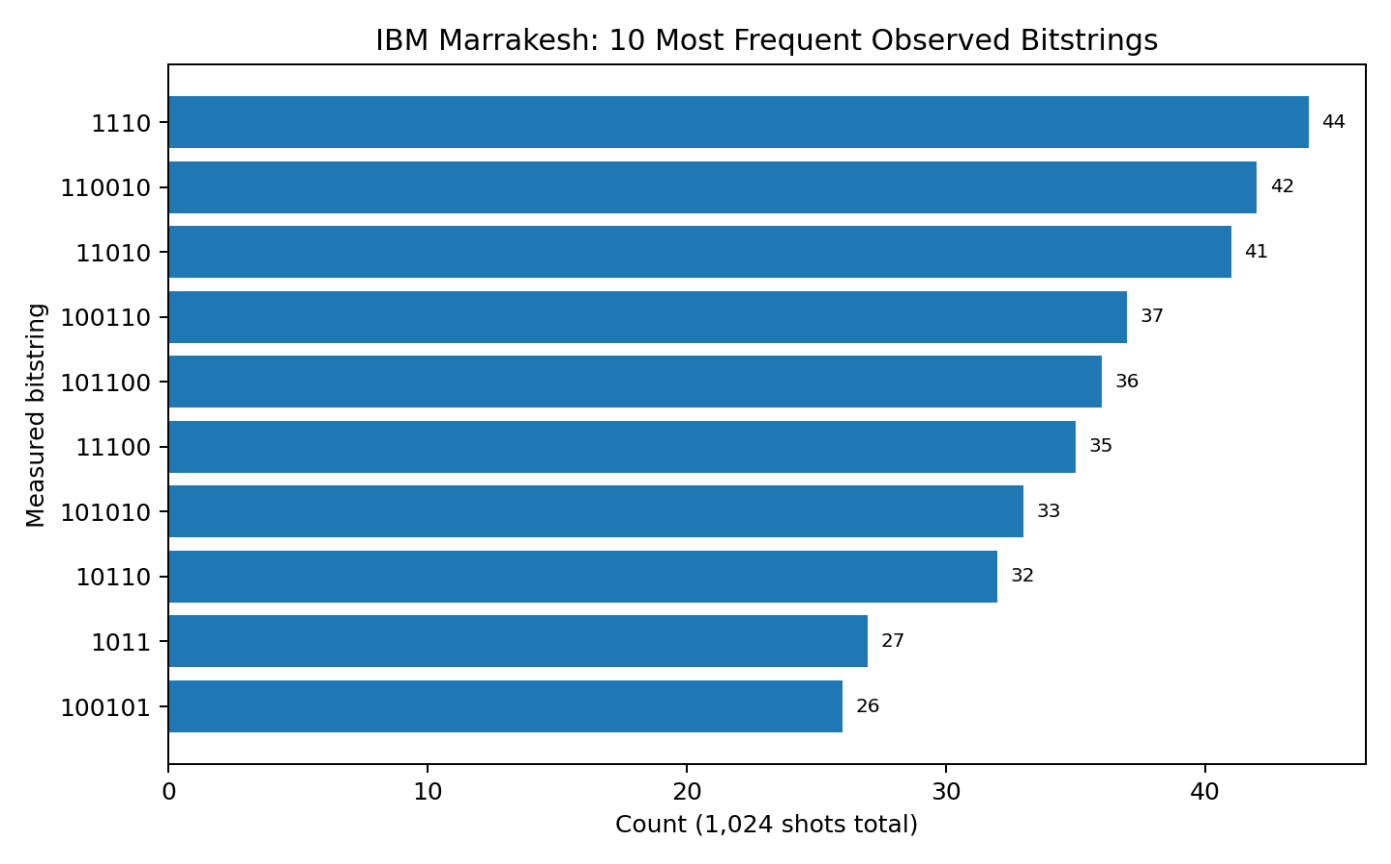
NOISY AER

7.47% feasible samples

This compares two simulations. It is not the same as the real IBM hardware run.

A completed IBM Quantum job returned 1,024 shots

Recorded run details and saved counts are included with the package.



JOB

ibm_marrakesh
ID: db83qs0r1c5c738lmabg

CIRCUIT

6 logical qubits
transpiled depth 144

OPTIMAL SAMPLE

100110 appeared 37 times
3.61% of shots

The bitstring matches the recorded exact optimum under the notebook's bit-order convention; this is not evidence of quantum advantage.

What each experiment can tell us

6 assets / QAOA	Simulator	Compare depth and penalty settings against exact optimum.
12 assets / QAOA	Simulator + exact search	Measure sampled quality against 924 feasible portfolios.
20 assets / 10 selected	Classical exhaustive search	Scale the classical constrained benchmark; not a QAOA result.
8-qubit pilot	Ideal + synthetic noisy Aer	Illustrate sensitivity to a noise model; not device-calibrated.
6-qubit IBM job	Real IBM Quantum Runtime run	Confirm completed execution and observed optimum sample; no advantage claim.

Separate labels matter: they distinguish measured device data from simulation and exact classical ground truth.

What the results do—and do not—show

SUPPORTED BY THE RECORD

Reproducible seeded instances, exact small-case references, saved simulator tables, and a completed IBM job with measurement counts.

NOT ESTABLISHED

Quantum advantage, a speedup over classical methods, device-calibrated noise conclusions, or generalization to real market portfolios.

Why this is useful: it is an end-to-end testbed for modeling, constraint handling, QAOA tuning, and evidence-based evaluation.

Next steps: repeat across seeds, compare more classical heuristics, calibrate noise assumptions, measure circuit/runtime costs, and document deployment behavior separately from algorithm results.

Open the Portfolio Optimizer

The deployed app is the interactive interface; the notebook and CSVs are the experiment record.

LIVE APP URL

https://qiskit-fall-fest-2026-iiitd-ajiq5qdibj2ujasptcycxm.streamlit.app/

Demo walkthrough

1. Open the app. 2. Inspect portfolio choices and constraints. 3. Compare the app's output with the notebook's seeded experiments and exact classical baselines. 4. Explain which results are simulated and which come from the recorded IBM job.

Optimizer deployment is external to this package; package checks verify the URL is embedded, not that the site is currently online.

The contribution is the comparison framework—not a ~~quan~~tum advantage claim

MODEL

Translate portfolio choices into binary variables and explicit constraints.

MEASURE

Track feasibility, cost, hit rate, shots, and circuit details.

COMPARE

Use exact classical ground truth and label each experiment honestly.

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Thank you